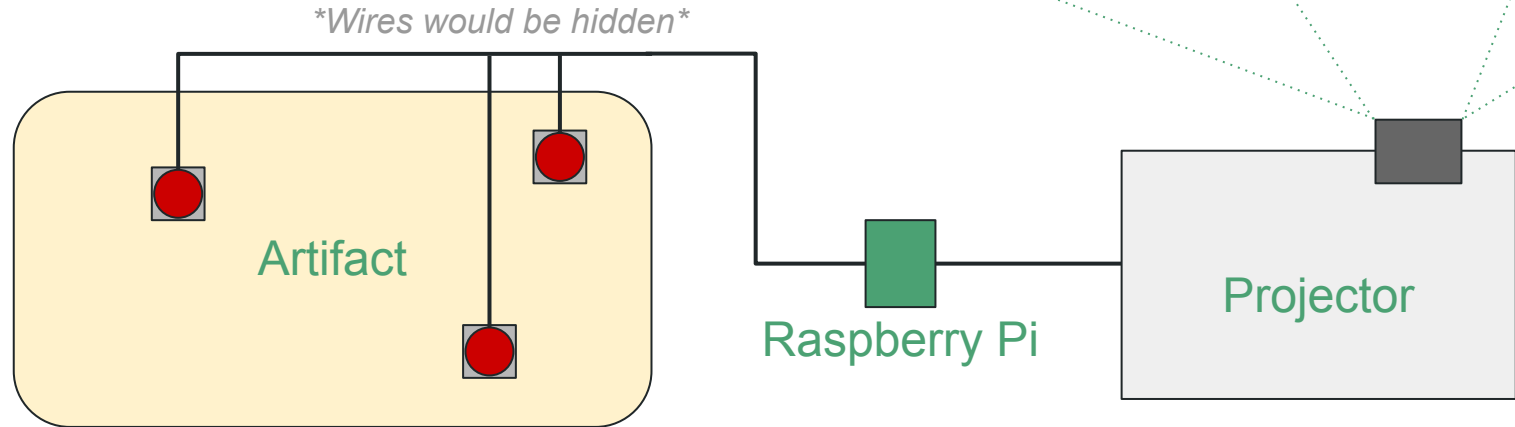


Wall Projections

Interactive Exhibitions for Bristol Museum

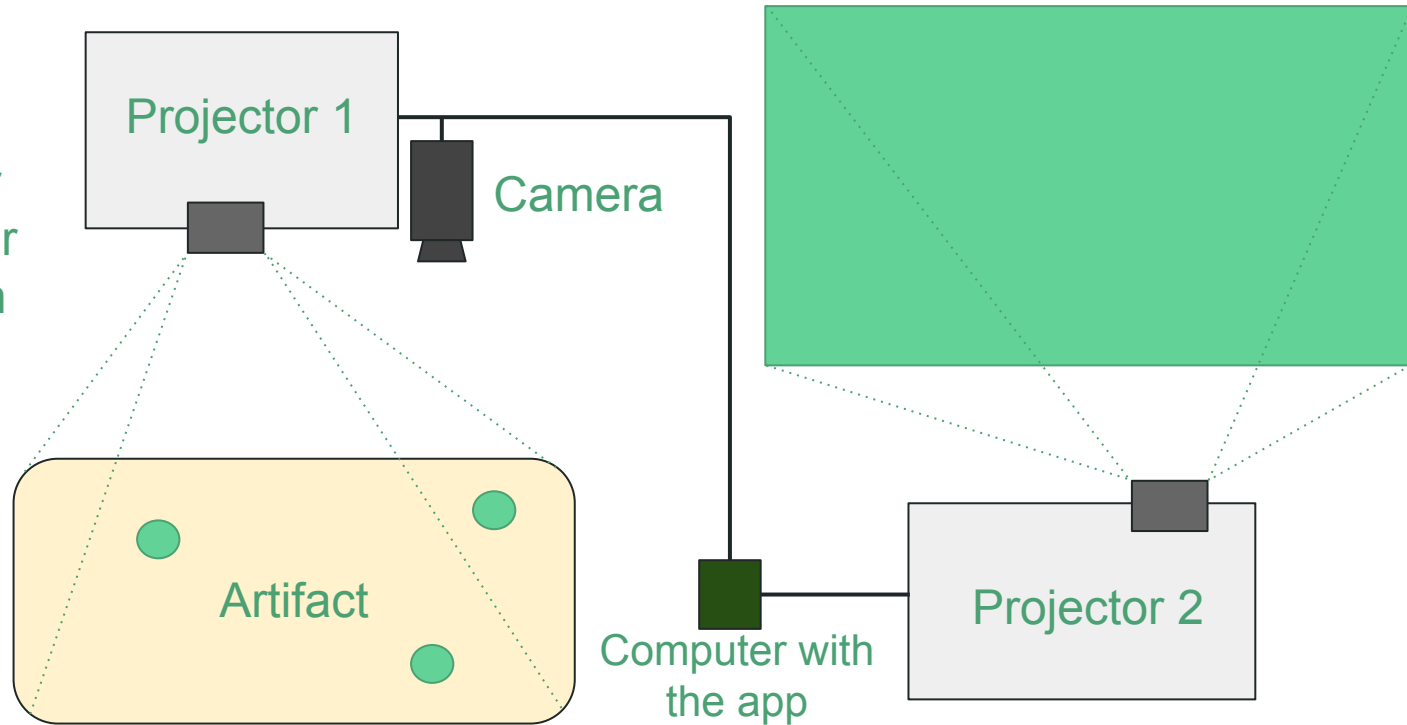
Minimal Viable Product

An artifact is set up with buttons placed at points of interest. When a user presses a button, information related to the point of interest is displayed on the Projector.

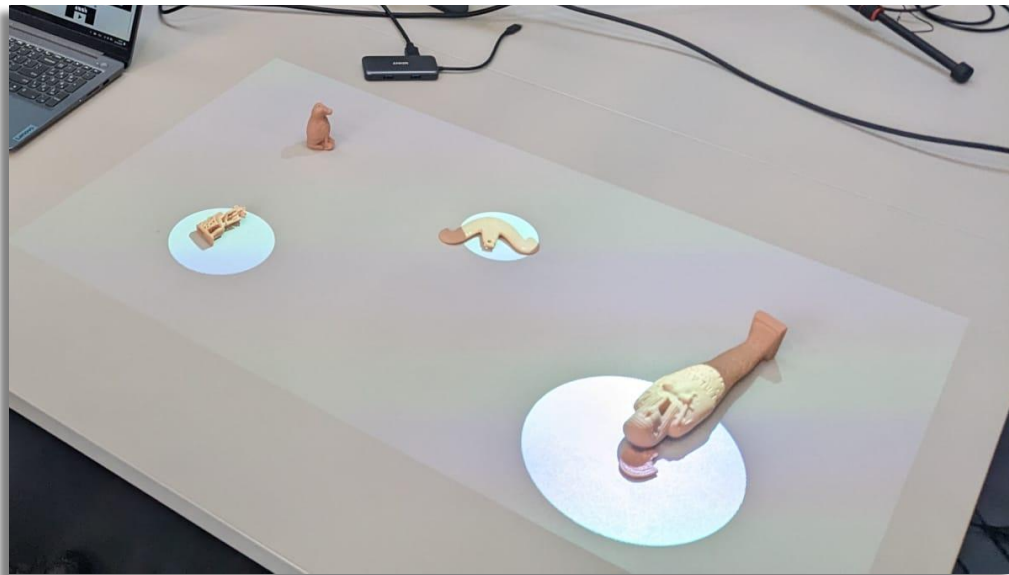


Beta release

An artifact is set up with hotspots projected at points of interest. When a user places their hand over a hotspot, information related to the hotspot is displayed on Projector 2.



Demo

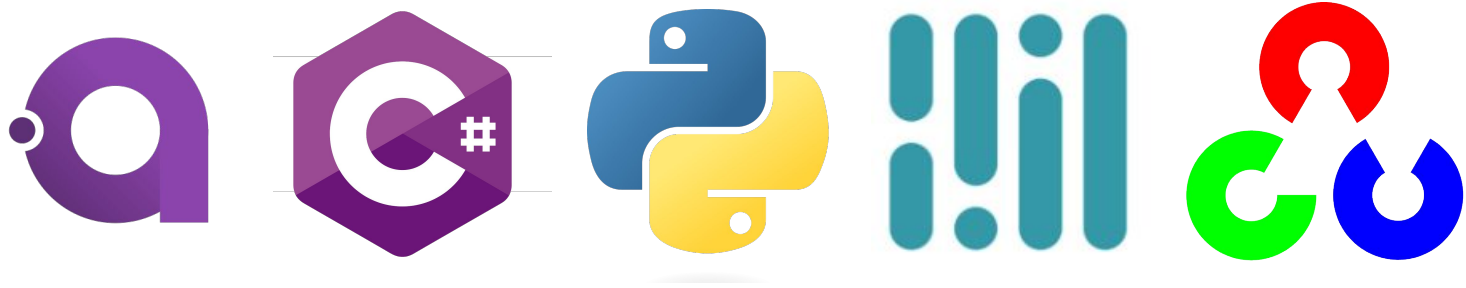


Some User Stories

- As a **museum visitor**, I want to interact with the installation, so that I can learn about the artefacts in an engaging way.
- As an **exhibit designer**, I want to edit hotspot information with a graphical interface, so that it is easier to line up the hotspots with the artefacts.
- As an **exhibit designer**, I want to edit the location of hotspots with a graphical interface, so that it is intuitive and takes less time.
- As an **exhibit designer**, I want to package the data, so that the museum curator can load it up without manual editing.

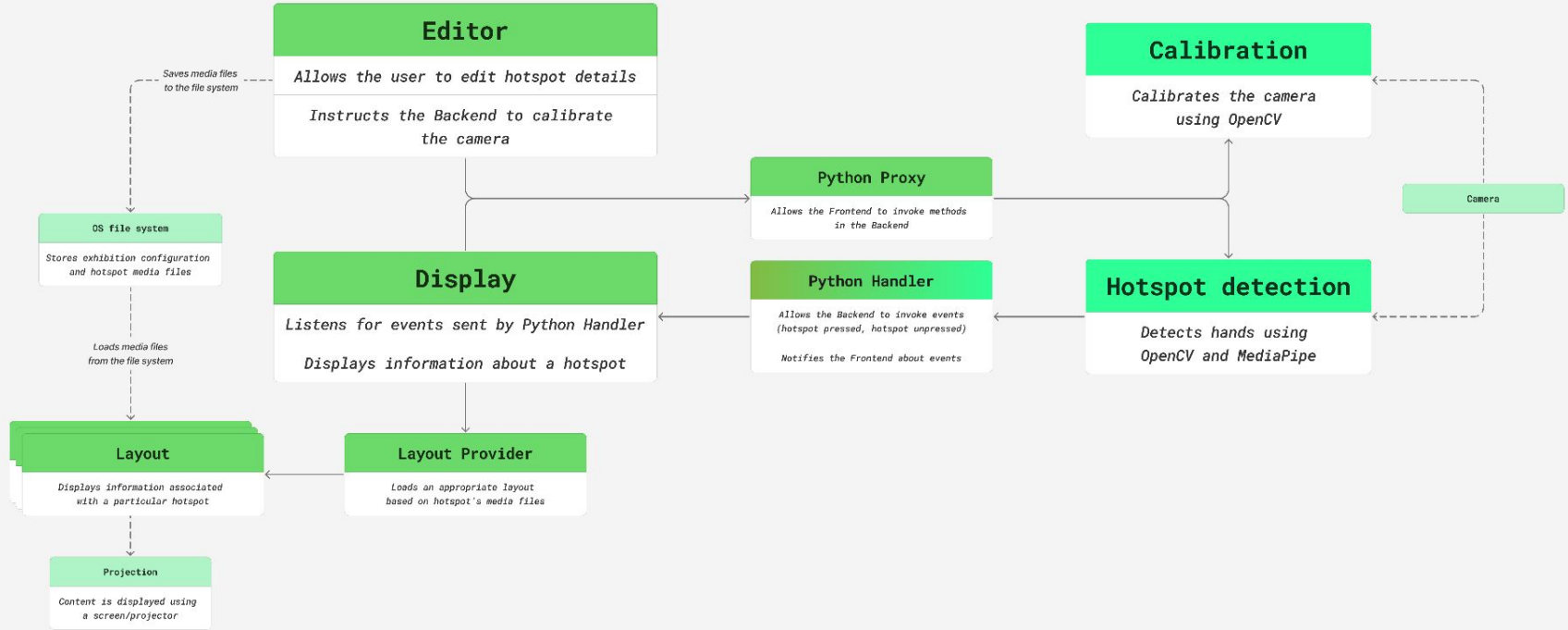
Technology Stack

Our project is written in 2 languages: *C#* for the Frontend and *Python* for the Backend. The former is built using *Avalonia*, a UI framework for *.Net* platform, and *python.net* library to communicate with the Backend. The latter uses *OpenCV* and Google's *MediaPipe* to detect hands on the camera feed.



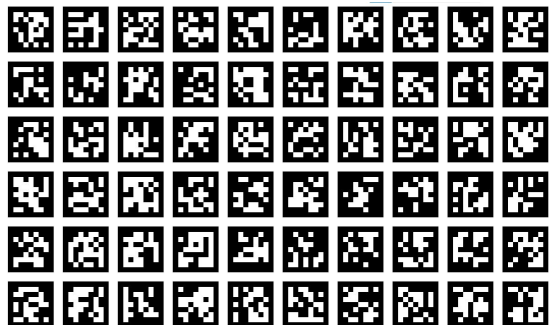
High-level System Architecture

Wall Projections - Beta

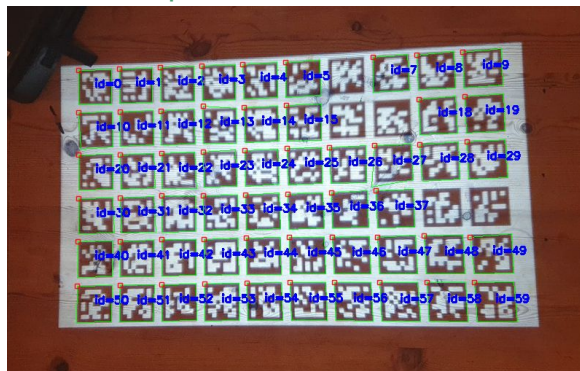


Computer Vision Calibration

Project a grid of ArUcos with known coordinates in “projector space”.



Detect coordinates of ArUcos in “camera space”.



Using the two sets of coordinates generate a Transformation Matrix from “camera space” to “projector space”.

$$\begin{bmatrix} 2.26026357 & 0.0140671786 & -174.477112 \\ 0.105387057 & 2.14734851 & -192.239216 \\ 0.000143228792 & 0.0000940191141 & 1 \end{bmatrix}$$

Computer Vision Hand Tracking

Record video of user interaction.



Downscale image.



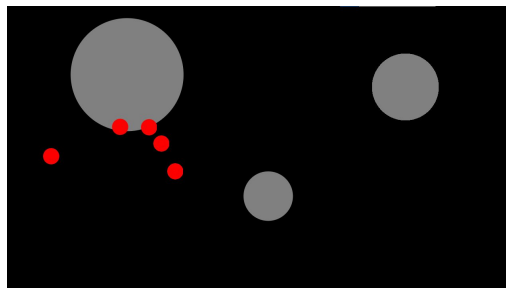
Run hand tracking model to identify landmarks.



Select for just fingertips.



Use the Transformation Matrix to transform fingertips to "projector space".



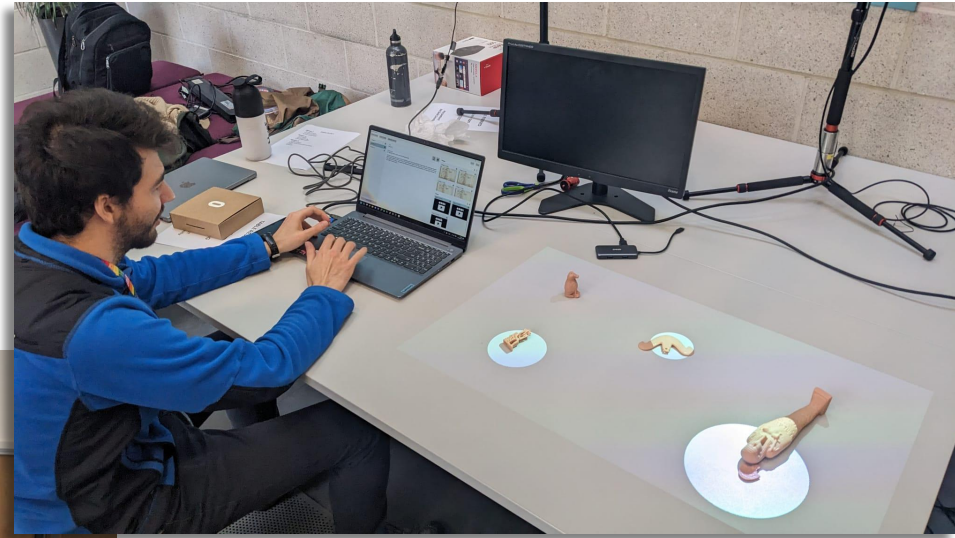
Check if fingertips are inside hotspots, communicate with frontend.

Backend development

- What are going to do? How are we going to do it?
 - Computer vision? ML? Lasers? Touch sensors? Light occlusion? Multiple cameras?
- Exploration stage
 - Exploring: Mediapipe, sensing decrease in light
- Hand landmarking proof-of-concept
 - Rough idea of scope and components of project
- Integration with frontend (ughhh)
 - Python-C# interop
- Calibration
 - Exploring: manual calibration, ArUco markers, ML-driven object detection
- More integration (aurghhh)
 - Physical set up and demo uncovered issues such as external camera

User testing

Photos from live demo on 15.03.2024



End goal

Things that we will work on for the final release:

- Make Editor's UI more intuitive
 - Most users had some trouble using certain features of the Editor
- Improve hotspot selection and animations
 - We want to add some delay before a hotspot is fully activated
- Make the Backend more robust and fault-tolerant
 - Discovering available cameras needs improving, in addition to general error recovery
- Test the system on Raspberry Pi 5
 - Raspberry Pi 4 proved to have insufficient computing power to display the information and run the Computer Vision at the same time

Questions